VU Quiz Collection

Total Questions: 17 from 17 Quiz(es)

Courses: MTH401

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Quiz 1 - MTH401 (Q1/10)

Question:

Recurrence relation for polynomials relates

Options:

A. Polynomials of different degrees.

B. Equal polynomials.

C. All options are true.

D. Polynomials of same degrees.

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Quiz 2 - MTH401 (Q2/10)

Question:

T h e D i f f e r e n t i a l E q u a t i o n (x 2" 1 6) y 2 2 + 2 x y 2 + 1 6 y = 0 (x 2" 1 6) y 3 + 2 x y 2 + 1 6 y = 0
(x^2 - 16)y'' + 2xy' + 16y = 0

has singularity at

Options:

A. x=16x=16
x = 16

B. None of these.

C. x=±4x=±4
x = \pm 4

D . x =" 1 6 x =" 1 6
x = - 16

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Quiz 3 - MTH401 (Q3/10)

Question:

A function f is said to be analytic at a point ‘a’ if it can be represented by a power series in (x-a) with a _____ radius of convergence.

Options:

A. negative

B. positive

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Quiz 4 - MTH401 (Q10/10)

Question:

The differential equation y22+(cosx)y=0y3+(cosax)y=0y'' + (\cos x)y = 0 has

Options:

A. x=0

B. x=1

C. x=-1

D. none of them

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Quiz 5 - MTH401 (Q5/10)

Question:

, and

Options:

A . x 1 2 = x 2 , x 2 2 = x 3 , x 3 2 = 1 4 x 1 + 1 4 e t x 1 2 = x 2 , x 2 2 = x 3 , x 3 2 = 1 4 x 1 + 1 4 e t { x _ 1 } ^ \ p r i m e {x_3},{x_3}^\prime = \frac{1}{4}{x_1} + \frac{1}{4}{e^t}

B. x12=x2,x22=x3,x32=" 14x1" 14etx12=x2,x22=x3,x32=" 14x1" 14et{x_1}^\prime = {x_2},{x_2} {x_3},{x_3}^\prime = - \frac{1}{4}{x_1} - \frac{1}{4}{e^t}

C. x12=x2,x22=x3,x32=" 14x1+14etx12=x2,x22=x3,x32=" 14x1+14et{x_1}^\prime = {x_2},{x_2}^\p 1}}{4}{x_1} + \frac{1}{4}{e^t}

D. x12=x2,x22=x3,x32=14x1" 14etx12=x2,x22=x3,x32=14x1" 14et{x_1}^\prime = {x_2},{x_2}^\pri \frac{1}{4}{x_1} - \frac{1}{4}{e^t}

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Quiz 6 - MTH401 (Q6/10)

Question:

If n is positive integer, then the solution of Legendre's differential equation is called a Legendre's _____ of degree n.

Options:

A. polynomial

B. function

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Quiz 7 - MTH401 (Q7/10)

Question:

and

Options:

A. x12=x2,x22=2x2" 52x1x12=x2,x22=2x2" 52x1{x_1}^\prime = {x_2},{x_2}^\p

B. x12=x2,x22=" 2x2+52x1x12=x2,x22=" 2x2+52x1{x_1}^\prime = {x_2},{x_2}^ \frac{5}{2}{x_1}

C. x12=x2,x22=2x2+52x1x12=x2,x22=2x2+52x1{x_1}^\prime = {x_2},{x_2}^\p

D. x12=x2,x22=" 2x2" 52x1x12=x2,x22=" 2x2" 52x1{x_1}^\prime = {x_2},{x_2}^ \frac{5}{2}{x_1}

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Quiz 8 - MTH401 (Q8/10)

Question:

A second order differential equation of the form
x 2 d 2 y d x 2 + x d y d x + (x 2" v 2) y = 0 x 2 d 2 y d x 2 + x d y d x + (x 2" v 2) y = 0 { x ^ 2 } \ f r a c { { { d ^ 2 } y } } { { d {v^2})y = 0 is called _____ differential equation.

Options:

A. Bessel

B. Picard

C. Legendre

D. Bernoulli

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Quiz 9 - MTH401 (Q9/10)

Question:

Operator method or systematic elimination method of solution of a system of linear homogeneous or linear non-homogeneous differential equations is based on the process of systematic elimination of the _____ variables.

Options:

A. independent

B. dependent

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Quiz 10 - MTH401 (Q10/10)

Question:

I f d e t e r m i n a n t "#"#"# D + 1 0 2 D" 1"#"#"# = 0 | D + 1 0 2 D" 1 | = 0 \ l e f t | { \ b e g i n { a r r a y } { * { 2 0 } { {\begin{array}{*{20}{c}}

{D + 1} \\\
0
\end{array}}&{\begin{array}{*{20}{c}}
2 \\\
{D - 1}
\end{array}}
\end{array}} \right| = 0 then _____.

Options:

A. D 2" 1 D 2" 1 { D ^ 2 } - 1

B. (D + 1) (D" 1)" 2 (D + 1) (D" 1)" 2 (D + 1) (D - 1) - 2

C. D2+1D2+1{D^2} + 1

D. (D + 1) (D" 1) + 2 (D + 1) (D" 1) + 2 (D + 1) (D - 1) + 2

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Quiz 11 - MTH401 (Q2/10)

Question:

A singular point x=x0x=x0x = {x_0} of the differential equation
a 2 (x) y 2 2 + a 1 (x) y 2 + a 0 (x) y = 0 a 2 (x) y 3 + a 1 (x) y 2 + a 0 (x) y = 0 { a _ 2 } (x) { y ^ { ' ' } } + { a _ 1 } (x
_ _ _ _ _ _ _ _ _ _ point if both (x" x0)P(x)(x" x0)P(x)(x - {x_0})P(x) and (x" x0)2Q(x
analytical at x0x0{x_0} .

Options:

A. ordinary

B. none of them

C. irregular singular

D. regular singular

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Quiz 12 - MTH401 (Q3/10)

Question:

Bessel functions of index half an _____ integer are called Spherical Bessel functions.

Options:

A. even

B. odd

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Quiz 13 - MTH401 (Q4/10)

Question:

The irregular singular point of the differential equation (x2" 4)2y+(x" 2)y2+y=0 (x - 2)y' + y = 0 is _____.

Options:

A. -2

B. -1

C. 2

D. 1

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Quiz 14 - MTH401 (Q5/10)

Question:

The differential equation 5dxdt+5x" dydt=e2t5dxdt+5x" dydt=e2t5\frac{{dx}}{{dt terms of differential operator is _____.

Options:

A. (D+1)5x" Dy=e2t(D+1)5x" Dy=e2t(D + 1)5x - Dy = {e^{2t}}

B. (D" 1)5x" Dy=e2t(D" 1)5x" Dy=e2t(D - 1)5x - Dy = {e^{2t}}

C. (D+1)5x" Dy=et(D+1)5x" Dy=et(D + 1)5x - Dy = {e^t}

D. (D+1)5x+Dy=e2t(D+1)5x+Dy=e2t(D + 1)5x + Dy = {e^{2t}}

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Quiz 15 - MTH401 (Q6/10)

Question:

The regular singular point of the differential equation (x2" 4)2y+(x" 2)y2+y=0((x - 2)y' + y = 0 is _____.

Options:

A. 2

B. 1

C. -1

D. -2

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Quiz 16 - MTH401 (Q7/10)

Question:

Every order differential equation as well as most systems of differential equations, could be reduced to the order system .

Options:

A. n" 1n" 1n - 1

B. n+1n+1n + 1

C. infinite

D. nthnth{n^{th}}

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Quiz 17 - MTH401 (Q9/10)

Question:

x=0 is a(an) -------- point of the differential equation:y22+xy2+2y=0.y3+xy2+2

Options:

A. Singular

B. Ordinary

C. Irregular singular

D. Regular singular

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